Toby Zhang

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Education

University of California, San Diego La Jolla, CA
B.S. Data Science, Minor in Cognitive Science: 3.8 GPA *Sept. 2024 – Present*

Relevant Coursework: Principles of Data Science, Intro to Programming, Linear Algebra, Calculus and Analytic Geometry For Engineering, Programming and Data Structures for Data Science, Intro to Probability, Intro to Research Methods, Data Structures and Algorithms for Data Science

Experience

Data Scientist January 2025 – Present
Data Science Student Society @ UCSD *La Jolla, CA*

- Collaborated on a time series analysis project to forecast economic trends for leading world powers, leveraging indicators like GDP, unemployment, inflation, CPI, and tariff data from World Bank and WTO datasets.
- Preprocessed raw economic data using Pandas to handle missing values, normalize datasets, and convert data into time series formats for model input.
- Created visualizations, including time series plots, correlation heatmaps, and comparative bar charts, with Matplotlib and Seaborn, to analyze and communicate insights on GDP trends, inflation rates, and trade impacts.

Projects

BingeRush – Personalized TV Show Recommendation System | *Python, Pandas, Flask* Apr. 2025

- Developed a content-based recommendation system using IMDb datasets (10M+ records), applying machine learning techniques to recommend TV shows by leveraging features like runtime, genres, ratings, and popularity
- Engineered features such as time closeness and normalized popularity, implementing a weighted scoring model (40% time closeness, 30% popularity, 20% rating, 10% episode count.) improving recommendation accuracy
- Deployed the system as a Flask web application with an HTML/CSS and Jinja2 frontend, enabling users to input preferences (e.g., genre, time limits) and receive real-time, data-driven binge-watching recommendations.
- Performed exploratory data analysis (EDA) to uncover trends in show popularity and runtime distributions, optimizing feature weights to prioritize discoverable content and demonstrating skills in data wrangling, statistical analysis, and user-focused solution design.

Python Multimedia Processing and KNN Classifier | *Python, NumPy, Pillow* Feb. 2025 - Mar 2025

- Developed and implemented image processing algorithms using Python and NumPy to manipulate RGB images, including transformations like negation, grayscale conversion, and edge detection.
- Designed and built a K-Nearest Neighbors (KNN) classifier for image classification, leveraging pixel-based Euclidean distance to improve predictive accuracy.
- Implemented multimedia processing functions such as audio reversal, speed adjustments, and reverberation effects using Python's wave and struct modules.

Patient Experience Sentiment Analysis | *TextBlob, Scikit-Learn, TdifVectorizer, Seaborn* Oct. 2024

- Performed feature engineering by cleaning and transforming patient survey data, handling missing values, and converting categorical data to numeric (e.g., HCAHPS answer percentages, sentiment scores)
- Extracted sentiment polarity from survey questions, classifying responses into Positive, Neutral, or Negative categories based on answer percentages.
- Trained a Logistic Regression model combining text (TF-IDF) and numeric features to predict sentiment categories
- Created visualizations with with Matplotlib and Seaborn. Visualized sentiment distribution with sample reviews, scatter plots weighted by survey respondents, and confusion matrix for model evaluation.

Technical Skills

Languages: Python, Java, HTML/CSS
Developer Tools: VS Code, Jupyter, Flask, Git, GitHub
Libraries: pandas, NumPy, Matplotlib, TextBlob, Scikit-Learn, TdifVectorizer, Seaborn